



Clinical Practice Guideline

High Flow Therapy (Humidified) (AIRVO2®)

Scope

Site	Service/Department/Unit	Disciplines
SCGH	Hospital-Wide	Medical, Nursing, Allied Health

Contents

Scope.....	1
Introduction	2
Risk Statement	2
Definitions	2
Key Points.....	3
Procedure	3
1. Indications for High Flow Therapy	3
2. Considerations for High Flow Therapy	3
3. Contraindications for High Flow Therapy	4
4. Prescribing	4
5. Application.....	4
5.1 Equipment Required	4
5.2 Set up.....	5
5.3 Titrating Therapy.....	6
5.4 Monitoring.....	6
5.5 Escalation	6
5.6 Additional Information	7
5.7 Transporting patients on High Flow Therapy.....	7
5.8 Weaning Therapy	7
5.9 Cleaning equipment.....	8
Appendix.....	10
1.1. Example of HFNP/CPAP/NIV Management & Prescription Chart (MR876)	10
1.2. Example of ORC modification in conjunction with HFT Prescription (MR876).....	10

Introduction

High Flow Therapy (HFT) administered via the AIRVO2® provides respiratory support by delivering heated and humidified air, **with or without** oxygen, at higher flow rates than conventional delivery methods.

It delivers a measured level of oxygen called the fraction of inspired oxygen concentrations (FiO₂) between 21%-100%

It is used with a patient interface of either high flow nasal cannula, tracheostomy connector or via a face mask adaptor.

The AIRVO2® can deliver humidified flow rates of 10 – 60L per minute that can match or even exceed the patients' inspiratory flow demand.

The increased flow rates provide benefits that oxygen alone cannot, such as reduction of dead space, and dynamic, positive airway pressure, which in turn can increase alveolar ventilation.

It is important to note that although HFT is a form of non-invasive respiratory support, it is NOT a ventilator or a CPAP device and does not replace the need for other NIV therapies such as BIPAP/CPAP.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGOPHCG Mission & Values	☒	Staff Safety	☒
Other:	☐		

Definitions

PO₂/PaO₂	Arterial partial pressure of oxygen obtained from an Arterial Blood Gas
ABG	Arterial Blood Gas
PCO₂/PaCO₂	Arterial partial pressure of carbon dioxide obtained from an ABG.
CBG	Capillary Blood Gas (G54 only) measure pH and PCO ₂
FiO₂	Fraction of inspired oxygen is the percentage concentration of the gas entering the lungs. This is calculated by the AIRVO2®
Flow	Refers to the total output rate (L/min) of room air and/or supplementary oxygen. This is set on the AIRVO2® machine as prescribed by Medical Officer
Oxygen Flow (L/m)	Refers to the oxygen flow rate added to the AIRVO2® via the side port

Key Points

- AIRVO2® can be used for airway humidification **with or without** oxygen.
- High Flow Therapy using the AIRVO2® can be used to deliver oxygen therapy accurately, assist airway clearance and to support hypoxic respiratory failure.
- Close monitoring is required if patients have respiratory failure as they may appear comfortable on AIRVO2® but have little respiratory reserve.
- HFT orders must be prescribed on the HFNO/CPAP/NIV management chart (MR- 876), modifications on ORC and oxygen prescription sticker
- A Medical review is recommended once patient reaches FiO₂ of 40% on 40L/flow
- An ICU review is recommended for patients requiring FiO₂ of ≥50% on a flow ≥40L, if this aligns with patient's goals of care

Procedure

1. Indications for High Flow Therapy

Hypoxic respiratory failure (arterial blood gas: PaO₂ < 60mmHg with a normal PaCO₂) eg:

- Acute Respiratory Distress Syndrome (ARDS),
- Pneumonia,
- Asthma
- Pulmonary Fibrosis
- post-operative hypoxemia
- Pulmonary Embolism
- Chronic Obstructive pulmonary disease (COPD) exacerbations with hypoxic respiratory failure (: PaO₂ < 60mmHg with a normal PaCO₂)
- NIV/CPAP weaning
- Influenza like illness (ILI) Note HFT is an aerosol generating procedure, use appropriate PPE (Personal Protective Equipment) + HEPA filter as per infection control guidelines.
- Preoxygenation prior to intubation
- Post extubation in low risk extubation failure
- 'Do not intubate'
- Palliative setting for symptom control
- Acute hypoxemic respiratory failure in immunocompromised patients
- Airway humidification

2. Considerations for High Flow Therapy

HFT can be used with caution and careful consideration with guidance from a respiratory, emergency, or critical care consultant in patients with:

- COPD and Hypercapnic Respiratory failure who do not tolerate NIV(BIPAP/CPAP)
- Cystic Fibrosis
- Obesity hypoventilation syndrome
- Neuromuscular disease e.g.: Guillian Barre Syndrome/ Motor Neuron Disease
- Patients with Chest wall deformities
- **SEVERE** Asthma
- Pneumothorax **ONLY** WITH Intercostal Catheter (ICC) insitu

3. Contraindications for High Flow Therapy

- Mid-maxillary facial trauma
- Base of skull fracture
- Surgery to nose or upper aero-digestive tract
- Gastric Distension
- Pneumothorax WITHOUT ICC
- Penetrating Lung Injuries
- Nasal Obstruction

4. Prescribing

HFT must be prescribed by senior medical personnel on the HFNO/CPAP/NIV management chart (MR 876). **(Appendix 1.1)** Prescription should include:

- Upper limit FiO₂ to trigger a medical review if patients SpO₂ falls below prescribed range
- Target SpO₂ % range: usually 94-98% in most patients or 88-92% in those patients at risk of or known CO₂ retention (COPD, chronic lung disease, obesity)
- Flow range e.g., 30-60L/min. Use lowest flow rate to maintain prescribed parameters
- Patients commenced on HFT therapy must have modifications on the Observation and Response Chart (MR-871) **(Appendix 1.2)** which align with the parameters prescribed on HFNO/CPAP/NIV management chart (MR 876)
- HFT must also be prescribed on Oxygen Prescription Sticker in medication chart
- An ABG is recommended prior to commencement of HFT in the setting of respiratory failure **(Appendix 2)**

For general ward care an upper FiO₂ limit of 40% and a flow rate of 40L is recommended to trigger a medical review.

If the goal of HFT/ AIRVO2® 2 is humidification, commence at 21%FiO₂ and flow of 30L/min.

5. Application

5.1 Equipment Required

- AIRVO2® unit
- AIRVO2® tube and chamber kit
- Patient interface (nasal cannula- available in S/M, tracheostomy connector)
- 1 Litre Water for Injection fluid bag
- Oxygen tubing

AIRVO 2 AND ACCESSORIES



Diagram 1

5.2 Set up

- Remove Red disinfection tubing (**do not discard, keep with the AIRVO2®**)
- Slide water chamber on to heat plate ensuring chamber ports inserted
- Pierce fluid bag to fill water chamber
- Connect heated breathing tube to port on top of AIRVO2®
- Connect delivery device i.e. nasal cannula to heated breathing tube
- Attach oxygen to side port if oxygen is required (Do not turn the oxygen on until Airvo unit is switched on)



- Turn on AIRVO2® and ensure the traffic light is green indicating the device has been disinfected since its last use.
- Hold the up and down arrows together for 3 seconds to unlock the screen. Use the mode button to scroll between features and to select desired settings.
- Set target dew point temperature to 37° for optimal humidification. If patient is unable to tolerate, reduce to 34°
- Set flow to 30L or prescribed flow rate.
- Apply nasal prongs/other interface, offer reassurance to patient and secure to patients gown with clip once patient is settled and comfortable

5.3 Titrating Therapy

- If oxygen is required, commence at 30% FiO₂ by adjusting the oxygen flow at the wall
- Assess the patients response by monitoring the respiratory rate and SpO₂
- Titrate oxygen as required to target the prescribed SpO₂
- If target SpO₂ is not achieved, consider increasing the flow rate prior to increasing FiO₂. Increasing flow may improve oxygenation, ventilation perfusion matching, reduce work of breathing and respiratory rate without the need for increased O₂ alone
- If little or no improvement in patient condition from increasing the flow, then FiO₂ should be titrated by adjusting the oxygen flow at the wall, within the prescribed limit of FiO₂ and oxygen saturations
- Patients failing to maintain adequate SpO₂ within the prescribed FiO₂ and flow must be referred for medical review.

When flow rates are increased, with no change to O₂ flow rate, the FiO₂ will decrease as O₂ is being diluted with air. Prescribed FiO₂ and oxygen saturations **MUST** still be targeted, which may require also increasing oxygen.

5.4 Monitoring

- When HFT is applied early enough, improvement in oxygenation, respiratory rate and dyspnoea should be seen within 15-30mins.
- Patients commenced on AIRVO2® require close monitoring and reassurance
- Stay with patient during the first 15 mins of initiation (unless purpose is purely humidification)
- Document FiO₂, flow rate, respiratory rate, and SpO₂ on the HFNC/CPAP/NIV management chart (MR 876) hourly for the first 4 hours
- Repeat ABG/VBG/CBG (CBG on G54 only) 1 hour post initiation
- A minimum 4 hourly recording of respiratory observations on the HFNC/CPAP/NIV management chart to be maintained once patient is stable and established on HFT.
- Ensure 4 hourly core physiological observation are also recorded on the ORC chart
- Any change in parameters or settings require a return to hourly observations for 4 hours post alteration

5.5 Escalation

- Titrate FiO₂ to meet the patient's requirements up to the prescribed FiO₂ limit
- Patients failing to maintain adequate SpO₂ within the prescribed FiO₂ must be referred for medical review
- Patients not tolerating treatment must be escalated for medical review or MET
- Consideration may be given to reducing the flow rate if not tolerated

The maximum recommended settings in the general ward area are 40% FiO₂ and 40L flow unless documented end of life plan or valid modifications and escalation criteria are in place.

Patients with increasing FiO₂/Flow exceeding this must be escalated for medical review when approaching these settings.

It is recommended a FiO₂ of ≥50% on a flow ≥40L have an ICU review if this aligns with the patient's goals of care¹⁵

5.6 Additional Information

- Patients receiving AIRVO2® therapy are limited to movement around bedside as the unit must remain plugged in to power source and fixed to oxygen ports on the wall
- Change circuit, humidifier base and patient interface every 7 days. Place yellow “change by” sticker on the circuit and document when change is next due in nursing care plan
- Perform regular mouth and nasal care
- Check humidifier water level and temperature regularly and replace fluid bag when indicated.
- Consider skin integrity and apply prophylactic dressings to pressure areas i.e. cheeks and ears in high-risk patients

5.7 Transporting patients on High Flow Therapy

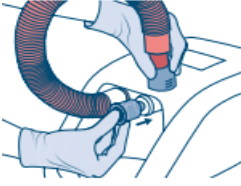
- The AIRVO2® does not have a battery therefore cannot be used during patient transfer
- The patient needs to be trialled on a transport oxygen delivery method for 10 minutes prior to transfer to ensure they can maintain oxygenation on alternative delivery device (**Appendix 4**)
- Standard oxygen delivery devices i.e. nasal prongs or mask are to be used as an alternative for transfer as per oxygen administration nursing practice guideline
- A nurse escort is required
- All consumables must be transferred with the patient if ongoing therapy required
- Ensure *Enterprise Bed Management* nursing needs are updated to reflect AIRVO2® requirement and contact the receiving ward prior to transfer

5.8 Weaning Therapy

- The decision to wean AIRVO2® treatment is made based on patient assessment and between senior nursing and allied health staff, in consultation with the medical team.
- When weaning, reduce FiO₂ by reducing oxygen prior to adjusting the AIRVO2® flow setting to promote more effective gas exchange while reducing O₂ requirements
- Decrease FiO₂ values to ≤35% by reducing oxygen, followed by a step-down decrease in flow rate using the AIRVO2® flow setting by 5-10L min until 15L/ min. When the patient is stable for 1-2hrs with FiO₂ 35% and flow ≤15L min HFT can be stopped, and conventional oxygen therapy can be used if desired.

5.9 Cleaning equipment

- After each patient's use, the AIRVO2® needs an external clean with combined detergent/disinfectant wipes e.g. Clinell wipes. Do not use wipe on left valve as can cause damage
- A new filter is to be inserted every 1000 hours; machine will alert (see diagram 1 for filter location)
- The red tubing is to be connected and disinfection cycle complete with new filter in situ (as below). Do not dispose of red tubing as this is reusable
- The red tubing should remain in situ to indicate clean machine ready for use. Only use if green light present as represents correct disinfection

	STEP 1: CONNECT THE RED DISINFECTION TUBE TO THE AIRVO One end of the disinfection tube has a blue plastic connector. Lift the sleeve and slide the connector onto the unit. Push the sleeve down to lock.
	Connect the red end of the disinfection tube to the left-hand chamber port as shown and push fully into position.
	STEP 2: FIT THE BLUE DISINFECTION FILTER Fit the blue disinfection filter to the right-hand chamber port as shown. Push it firmly into position.
	STEP 3: CONNECT THE POWER CORD Connect the power cable to the mains/utility power socket.
	STEP 4: TURN ON THE AIRVO DEVICE Start the disinfection cycle by holding the power On/Off button for 3 seconds until the unit beeps. The unit will display a warming up screen whilst performing a series of calibration checks.
	STEP 5: CHECK DISINFECTION CYCLE HAS STARTED After a successful calibration, the disinfection cycle begins and runs for 55 minutes, the remaining time displayed will count down to zero.
	STEP 6: CHECK DISINFECTION CYCLE HAS COMPLETED After a successful disinfection, the display will reach zero and an audible information signal will sound. The current disinfection cycle number will also be displayed. This number can be used for keeping a log of disinfection cycles.
	STEP 7: SWITCH OFF THE AIRVO Switch off the unit and unplug from the mains/utility power socket. The unit is now ready for storage.

Related Documents

[SCGOPHCG Oxygen Therapy CPG](#)

[SCGOPHCG IP&C Standard and Transmission-based Precautions Policy](#)

[SCGOHPCG Recognising and Responding to Acute Deterioration](#)

[Fisher & Paykel AIRVO2 page – including instructional videos](#)

[AIRVO2 User Manual](#)

[AIRVO2 Disinfection Kit Manual](#)

References

1. National COVID-19 Clinical Evidence Taskforce: <https://covid19evidence.net.au/>
2. AIRVO2® user manual. Fisher Paykel Healthcare. <https://resources.fphealthcare.com>
3. Frat J-P, Thille A, Mercat A et al. High-Flow Oxygen through Nasal Cannula in Acute Hypoxemic Respiratory failure. *New England Journal of Medicine* 2015; 372:2185-2196
4. Parke R, McGuinness S, Eccleston M. Nasal high-flow therapy delivers low level positive airway pressure. *Br J Anaesth.* 2009; 103(6): 886-90.
5. Rocca O, Riera J, Torres F et al. High-flow oxygen therapy in acute respiratory failure. *Resp Care.* 2010; 55(4): 408-13.
6. Pantazopoulos I, Daniil Z, Moylan M, Gourgoulis K, Chalkias A, Zakynthinos S, Ischaki E. Nasal High Flow Use in COPD Patients with Hypercapnic Respiratory Failure: Treatment Algorithm & Review of the Literature. *COPD.* 2020 Feb;17(1):101-111. doi: 10.1080/15412555.2020.1715361. Epub 2020 Jan 22. PMID: 31965868.
7. Raoof S, Nava S, Carpati C, Hill NS. High-Flow, Noninvasive Ventilation and Awake (Nonintubation) Prone in Patients With Coronavirus Disease 2019 With Respiratory Failure. *Chest.* 2020 Nov;158(5):1992-2002. doi: 10.1016/j.chest.2020.07.013. Epub 2020 Jul 15. PMID: 32681847; PMCID: PMC7362846.
8. Oczkowski S, Ergon B, Bos L, Chatwin M, Ferrer M, Gregoretti C, Heunks L, Frat JP, Longhini F, Nava S, Navalesi P, Ozsancak Ugurlu A, Pisani L, Renda T, Thille AW, Winck JC, Windisch W, Tonia T, Boyd J, Sotgiu G, Scala R. ERS clinical practice guidelines: high-flow nasal cannula in acute respiratory failure. *Eur Respir J.* 2022 Apr 14;59(4):2101574. doi: 10.1183/13993003.01574-2021. PMID: 34649974.
9. Piraino T, Madden M, Roberts KJ, Lamberti J, Ginier E, Strickland SL. AARC Clinical Practice Guideline: Management of Adult Patients With Oxygen in the Acute Care Setting. *Respir Care.* 2022 Jan;67(1):115-128. doi: 10.4187/respcare.09294. Epub 2021 Nov 2. PMID: 34728574.
10. Ischaki E, Pantazopoulos I, Zakynthinos S. Nasal high flow therapy: a novel treatment rather than a more expensive oxygen device. *Eur Respir Rev.* 2017 Aug 9;26(145):170028. doi: 10.1183/16000617.0028-2017. PMID: 28794144; PMCID: PMC9488954.
11. Barnett A, Beasley R, Buchan C, Chien J, Farah CS, King G, McDonald CF, Miller B, Munsif M, Psirides A, Reid L, Roberts M, Smallwood N, Smith S. Thoracic Society of Australia and New Zealand Position Statement on Acute Oxygen Use in Adults: 'Swimming between the flags'. *Respirology.* 2022 Apr;27(4):262-276. doi: 10.1111/resp.14218. Epub 2022 Feb 17. PMID: 35178831; PMCID: PMC9303673.
12. Espiney R, Jablonski J, Brys M, et al.: Humidified High Flow Nasal Cannula Algorithm for Primary Therapy in Hypercapnic Respiratory Failure. *Glob J Anes & Pain Med.* 2019;1:86–91. 10.32474/GJAPM.2019.01.000120
13. The Royal Perth Bentley Group clinical practice standard for Oxygen Therapy Management (for adult inpatients) 2022 version 5
14. Fiona Stanley Fremantle Hospital Group. Humidified high flow nasal cannula in adults Reference #: FSFH-WW-POL-0051 version 4 2022
15. Barnett A, Beasley R, Buchan C, et al. Thoracic Society of Australia and New Zealand Position Statement on Acute Oxygen Use in Adults: 'Swimming between the flags'. *Respirology.* 2022 Apr;27(4):262-276. doi: 10.1111/resp.14218. Epub 2022 Feb 17. PMID: 35178831; PMCID: PMC9303673.

Authorisation & Version Control

Policy Sponsor	Medical Co-Director Cancer, Imaging and Clinical Services				
Policy Contact	Clinical Nurse Specialist G54				
Date First Issued:	June 2023	Last Reviewed:	21/03/2024	Review Date:	21/03/2027
Version No.	2	Reference #:	CPG005		
Approved by:	Medical Co-Director Cancer, Imaging and Clinical Services			Date:	13/05/2024
Endorsed by:	SCGOPHCG Policy and Guideline Committee			Date:	
Acknowledgments:	Rebecca Stringer CNS, Chris Kosky Respiratory and Sleep Consultant, Rachel Hogg, Kirsty Officer CNS				
Standards Applicable					

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

Appendix 1

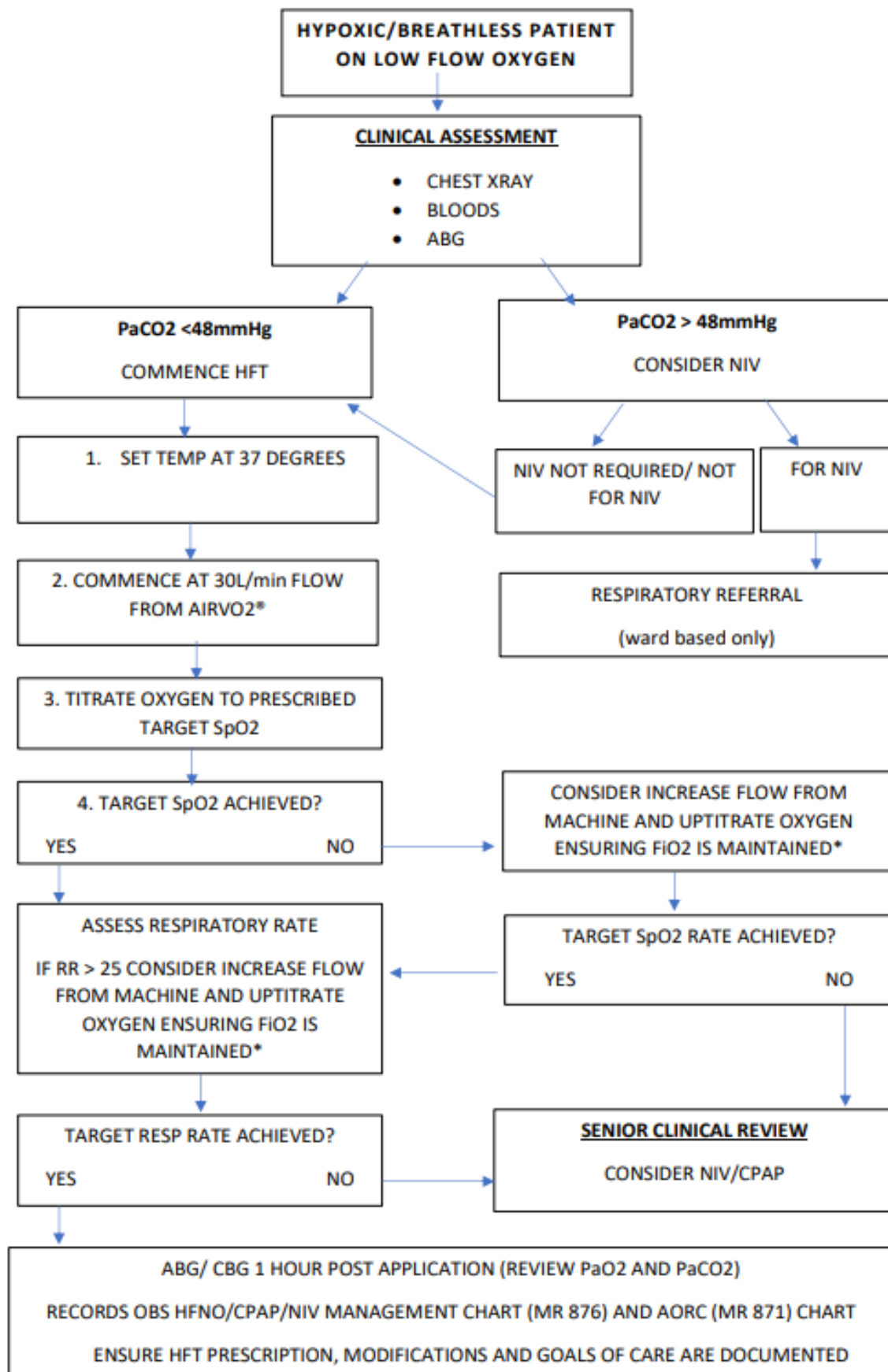
SIR CHARLES GAIRDNER OSBORNE PARK HOSPITAL GROUP		URN: A12345678		
HFNO / CPAP / NIV MANAGEMENT CHART		Family name: Bloggs		
		Given name(s): Joe		
		Gender:	DOB: 29/02/1956	
Terminology and Equations	EPAP	Expiratory Positive Airway Pressure	IPAP	Inspiratory Positive Airway Pressure
	PEEP	Peek End Expiratory Pressure	PS	Pressure Support
			Pinsp	Pressure Inspired (Hamilton T1)
	PEEP = EPAP		PS = Pinsp (for Hamilton T1)	
	PEEP + PS = IPAP		PEEP + Pinsp = IPAP	
	Duration E.g. continuous, while sleeping, while awake			
Initial High Flow Nasal Oxygen (HFNO) Order				
Date:	01/03/2024	Time:	14:00 hrs	
Flow Rate (L/min):	30-40L			
FiO ₂ :	21% - 40% if $\geq 40\%$ MR	Target SpO ₂ :	Aim 88-92%	
Doctor signature:	A.Smith <i>A.Smith</i>	Designation:	Snr Med Registrar	

1.1. Example of HFNP/CPAP/NIV Management & Prescription Chart (MR876)

SIR CHARLES GAIRDNER HOSPITAL		URN: A12345678	
OBSERVATION AND RESPONSE CHART		Surname: Bloggs	
		Forename: Joe	
		Gender: M	DOB: 29/02/1956
Other Charts in Use			
☐ Alcohol/Drug Withdrawal Chart ☐ Regional/Epidural/Patient Controlled Analgesia Chart ☐ Fluid Balance Chart ☐ Other _____ ☐ Neurovascular Chart			
Refer to above charts for additional instructions.			
Clinical Management Plan for Altered Parameters			
If adjusted parameter ranges are to be acceptable for the patient's clinical condition, the table below including range, escalation required along with rationale and justification must be completed.			
Observation	Parameter Range	Escalation Required (Please indicate: SNR / MR / MET)	
Respiratory Rate	-		
Oxygen Saturation	Aim 88-92%	86-88% SNR, <85% MR	
Systolic BP	-		
Heart Rate	-		
Temperature	-		
Other (specify)	-		
Clinical Rationale & Justification (Registrar/Consultant)			
Note: Modification WILL NOT be valid unless clinical rationale and justification are documented			
B/G Severe COPD risk of hypercapnic respiratory failure.			
See High flow prescription MR876			
Date / Time	Doctor's name	Doctor's signature	
01/03/2024	A.Smith	<i>A.Smith</i>	

1.2. Example of ORC modification in conjunction with HFT Prescription (MR876)

Appendix 2 Humidified High Flow Therapy (HFT) Flow Chart



*Increasing flow rates on AIRVO2® will reduce FIO2

Appendix 3: Flow capabilities of AIRVO2®

FiO2 capabilities of Airvo 2															
		O2 L/min (on wall)													
		1	3	5	7	10	15	20	25	30	35	40	50		
Flow Rate L/min (on machine)	20	25%	32%	40%	44%	59%	77%	95%	O2 too high						
	25	25%	30%	36%	42%	51%	66%	81%							95%
	30	25%	29%	34%	39%	46%	56%	71%							84%
	35	21%	27%	32%	36%	42%	53%	64%	75%	85%	95%				
	40	21%	27%	30%	34%	40%	49%	59%	68%	77%	87%			95%	
	45	21%	26%	29%	33%	38%	46%	54%	63%	71%	79%			88%	
	50	21%	26%	29%	32%	36%	44%	51%	59%	66%	74%	81%	95%		
	55	21%	25%	28%	31%	35%	41%	48%	55%	62%	69%	76%	89%		
	60	21%	25%	27%	30%	34%	40%	46%	52%	59%	65%	71%	89%		

Appendix 4: Portable oxygen interfaces FiO₂ capabilities

Oxygen deliver device	FiO ₂ capabilities
Nasal Cannula	Can deliver an FiO ₂ 24–35% at an oxygen flow of 1–4 L/min
Venturi Masks	Can deliver an FiO ₂ of 24–60%
Hudson Mask	Can deliver an FiO ₂ of 35–60% at an oxygen flow of 5-10 L/min.
Non-rebreather Mask and Bag	Can deliver an FiO ₂ of 60-90% at an oxygen flow rate of 10-15 L/min.